Panel Drilling
Quality Inspection Standard

DOCUMENT NUMBER	VERSION	EFFECTIVE DATE
YS-ZZZX-007	A/0	21 March 2026

Equipment: Five-sided and six-sided drilling machines

Operator requirements: Operators must be trained and qualified, and must be proficient in equipment operation, equipment performance, safety precautions and protective measures.

Pre-operation inspection

1. Check the incoming panel surfaces, cut dimensions and edge banding quality. Verify that the appearance after drilling meets the quality requirements.

Workmanship requirements

1. Hole edges must not be chipped or darkened. Remove wood chips from holes. The reverse face around a hole must not split or bulge.
2. Workpieces and their edges and corners must be free from pressure marks, scratches and impact damage. Edge banding must be undamaged. Panel surfaces must not be repaired; any exceptional case requires the manager's approval before release.
3. Hole-spacing deviation must be less than 0.3 mm and hole-depth deviation must be less than 0.5 mm, unless special requirements apply.
4. Shaped machining and all grooves must be free from edge chipping.
5. Position panels squarely for processing; do not load them at an angle.
6. Drill according to the label requirements. Do not miss or omit any holes.
7. Process small panels at the specified speed to prevent movement out of position.
8. Raise the lower stop before loading large panels.
9. Clarify any uncertainty in the drawings before processing. Do not make unauthorized assumptions.
10. Handle workpieces gently and do not place them directly on the floor.

Inspection and Release

Panel Drilling

Inspection and release requirements

1. Check mating hole positions by fitting the associated panels together.
2. Carry out self-inspection during processing and perform random inspections five times to verify product quality.
3. Regularly check surfaces, edges, corners and edge bands for machine scratches, pressure marks, abrasion and other damage.
4. Select the correct processing face and distinguish the quality requirements for the different panel faces.
5. After completing the operation, sign the Order Details Sheet. Transfer the complete set to the next operation only after confirmation.

Stacking and panel protection

1. Do not place processed panels directly on the floor. Place them on support boards and keep them within the support-board footprint.
2. Keep stacked panels clean and free from hard objects between panels to prevent surface scratches.
3. Stack panels no higher than 0.8 m.
4. Transfer complete sets to the designated lane for the next operation according to the panel processing route.

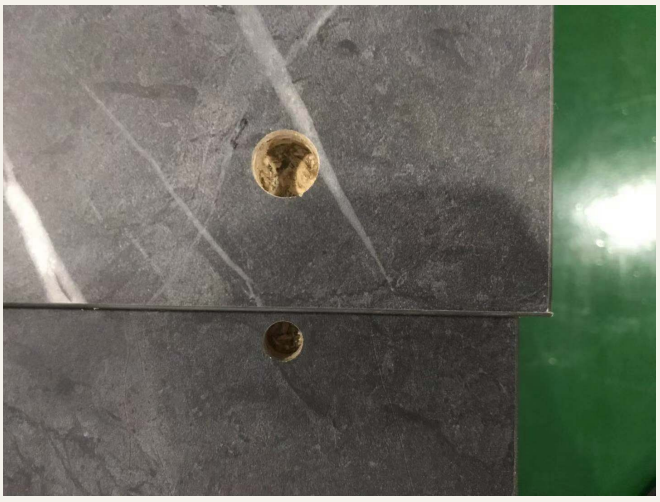
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Inspection Examples

Panel Drilling / Acceptance and defect references



ACCEPTABLE

Acceptable drilling



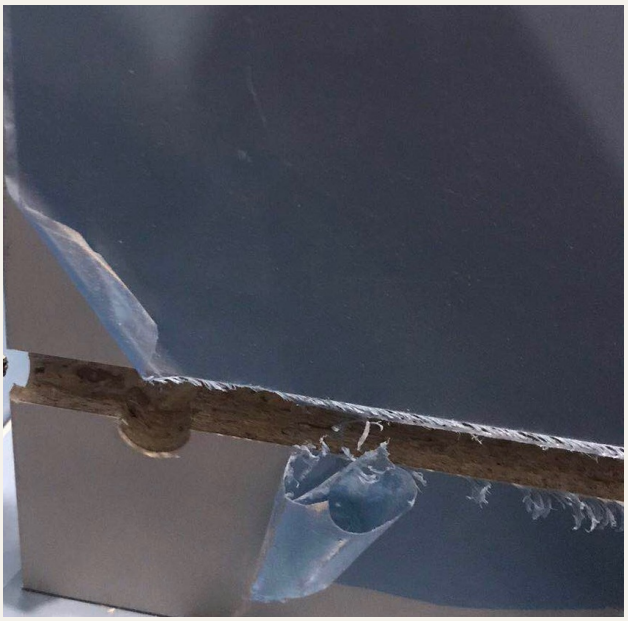
ACCEPTABLE

Acceptable drilling and grooving



NONCONFORMING

Hole position deviation

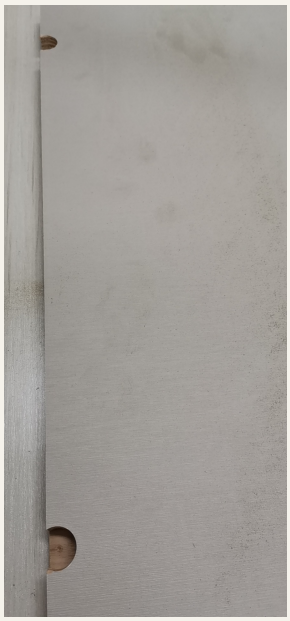


NONCONFORMING

Hole conflicts with the lighting groove

Inspection Examples

Panel Drilling / Acceptance and defect references



NONCONFORMING

Misaligned hole position



NONCONFORMING

Broken corner



NONCONFORMING

Chipped opening



NONCONFORMING

Intersecting holes

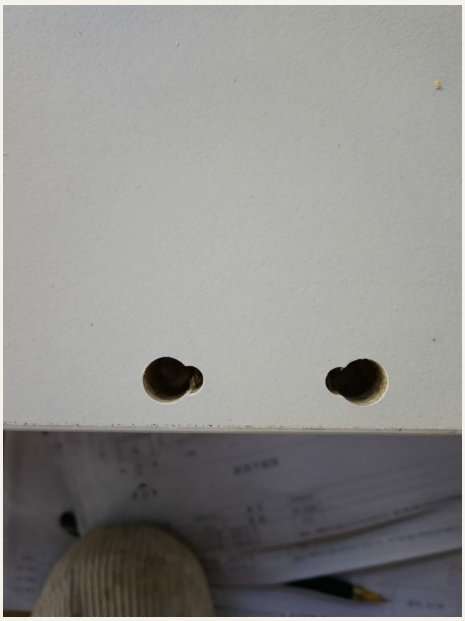
Inspection Examples

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NONCONFORMING

Groove too wide



NONCONFORMING

Incorrect hole position